

Advanced HDPE Pipe Technical Datasheet

This technical datasheet provides detailed engineering, performance, dimensional, pressure, installation, and compliance information for high-density polyethylene (HDPE) piping systems designed for industrial, irrigation, infrastructure, mining, water distribution, and gas transport applications.

1. General Product Information

Parameter	Value
Material Type	High Density Polyethylene (HDPE)
Resin Grade	PE63 / PE80 / PE100
Manufacturing Standard	IS 4984, ISO 4427
Pipe Color	Black with Blue Stripes
Application	Water Supply, Irrigation, Sewerage, Gas
Country of Origin	India
Expected Service Life	50+ Years
UV Resistance	Excellent
Corrosion Resistance	100% Corrosion Free

2. Dimensional Specifications

Parameter	Range
Outer Diameter	20mm – 1600mm
Wall Thickness	2.3mm – 118mm
Length Type	Coils / Straight Lengths
Straight Pipe Length	6m / 12m
Coil Length	50m – 500m
SDR Values	SDR 41, 33, 26, 21, 17, 13.6, 11
Ovality Tolerance	≤ 1.5%
Thickness Tolerance	± 0.3mm

3. Pressure Ratings

Pressure Class	Working Pressure
PN 2.5	2.5 Bar
PN 4	4 Bar
PN 6	6 Bar
PN 8	8 Bar
PN 10	10 Bar
PN 12.5	12.5 Bar
PN 16	16 Bar

4. Mechanical Properties

Property	Value
Density	0.95 – 0.96 g/cm3
Tensile Strength	≥ 22 MPa
Elongation at Break	> 350%
Flexural Modulus	800 – 1100 MPa
Impact Resistance	Excellent
Abrasion Resistance	Very High
Hardness	Shore D 60 – 68
Thermal Expansion	0.2 mm/m°C

5. Thermal Properties

Parameter	Value
Operating Temperature	-40°C to +80°C
Melting Point	130°C – 137°C
Thermal Conductivity	0.4 W/mK
Heat Stability	Excellent
Cold Weather Performance	High

6. Chemical Resistance

Chemical Type	Resistance
Acids	Excellent
Alkalies	Excellent
Salt Water	Excellent
Industrial Chemicals	Very Good
Alcohols	Excellent
Hydrocarbons	Moderate

7. Jointing Methods

Method	Description
Butt Fusion	Permanent thermally fused joint
Electrofusion	Electrical fitting-based fusion
Compression Fittings	Easy installation
Flange Joint	Industrial equipment connection
Mechanical Coupling	Repair and maintenance

8. Installation Guidelines

Requirement	Recommendation
Trench Width	Pipe OD + 300mm
Bedding Material	Fine sand / granular soil
Minimum Cover	750mm
Hydrostatic Testing	1.5 × Working Pressure
Bending Radius	20 × Pipe OD
Storage	Away from sharp objects

9. Compliance & Certifications

Standard	Description
IS 4984	HDPE pipes for water supply
ISO 4427	Plastic piping systems
ASTM D3035	Polyethylene piping
ISO 9001	Quality Management
WRAS Approved	Potable water applications

This document is a sample engineering datasheet created for website demonstration purposes. Actual product specifications may vary depending on manufacturer, application, and standards.